A Spatial Reasoning Benchmark for Multimodal Language Models

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1 Introduction

Spatial reasoning refers to the ability to understand how objects exist and change within space. It involves recognizing an object's position, orientation, and movement, as well as understanding how those properties change when the object is moved or viewed from a different perspective. People rely on this type of reasoning frequently in everyday situations. It appears when navigating through a building, interpreting diagrams, or imagining how an object would look after being turned or repositioned. Many puzzles and visual problems depend on the same type of thinking, where the difficulty comes from mentally tracking how an object changes while maintaining an accurate understanding of its orientation and position. Despite the rapid progress of multimodal language models, spatial reasoning remains an area where performance can still be inconsistent. Models often perform well when answers can be inferred from patterns in training data, but may struggle when the task requires maintaining a consistent understanding of where an object is as it moves through space. Small variations in wording or orientation can sometimes lead to different answers even when the reasoning process should remain the same. This possibly means that models may depend on pattern recognition more than making a representation of the spatial relationship of the object and its surroundings. This proposal introduces a benchmark designed to evaluate spatial reasoning in both text-only and multimodal settings. The goal is to measure whether a model can maintain a consistent representation of an object as it is moved in space. The tasks are structured around sequences of spatial operations where an object begins in a default position and then goes through one or more transformations. These problems resemble the type of reasoning used when solving spatial puzzles, where the challenge lies in tracking the transformation rather than identifying the object itself. The benchmark will include a variety of tasks that test the same spatial reasoning ability throughout different formats. Some questions will be presented entirely through text descriptions, while others will include simple images that represent an object and its orientation. Each task will require the model to follow a sequence of spatial changes and determine the final orientation or relative position of the object. By constraining the answer space and varying the transformations used, the benchmark aims to isolate spatial reasoning ability rather than allowing models to rely on memorized patterns. The tasks focus on common spatial transformations such as rotations, directional movement,

and relative positioning, requiring the model to maintain a consistent internal representation of an object's orientation and location across multiple steps.

2 Concept & Operationalization

This benchmark will have several exercises to measure LLM's ability to use common-sense spatial reasoning. We define common-sense reasoning as the ability to correctly infer relative orientation, rotation, and spatial displacement of objects under geometric transformations, using either textual descriptions alone or text grounded in visual input. This includes; mental rotation, relative positioning, directional movement inference, and transformation composition.

- *Mental transformation* is the ability to infer an object orientation after rotation, sequential composition of transformation. This includes both single-step transformations (e.g., rotate 90ř clockwise) and sequential composition of transformations (e.g., rotate 90ř clockwise and then 180ř). Correctly completing these types of exercises requires maintaining an internal representation of orientation and correctly updating it under rule-based operations.
- *Relative positioning* is the ability to define an object's spatial location with respect to another object or reference point. For example, in the statement "The ball is to the left of the chair.", the model must correctly interpret directional relationships within a shared coordinate frame. This includes reasoning about object-centered versus global reference frames and maintaining consistency when the reference object changes.
- *Directional movement* is the ability to interpret and apply movement commands, such as up, forward, clockwise, or counterclockwise. These tasks require mapping keywords to geometric operations and updating positional states accordingly. Movement reasoning may involve translating a point across a grid, following the direction of an arrow, or applying rotation to a facing object.
- *Transformation composition* is the ability to apply multiple spatial transformations sequentially to one object and identify its final state. This tests whether the models are performing true compositional reasoning or relying on shallow pattern recognition. For example, combining rotation and translation requires maintaining intermediate state representations and applying transformation rules in order.

Including these four categories of capabilities will evaluate whether a model maintains internally consistent spatial representations that are stable across textual and visual modalities. Our tasks for this benchmark will be split into two types of exercises; text-only spatial reasoning tasks and image + text multimodal tasks. Each task will have a standardized input-output structure, meaning that all tasks will follow a uniform prompt template. Each task will explicitly instruct the model to select a single-word answer from a predefined answer bank. This benchmark will test models performance on these topics using various exercises. We will construct it as a controlled benchmark composed of discrete tasks across multimodal settings. Each task will evaluate a specific skill under the large umbrella of spatial reasoning. By using this benchmark, we will evaluate how effectively large language models understand and manipulate physical space layouts, and object relationships.

3 Related Benchmarks

Other benchmarks have previously tried measuring LLM's consistency when it comes to true spatial intelligence/awareness. A model is able to detect a car in a photo, but it often cannot truly "think" about how that car would interact within a three dimensional context. What would a car look like if it was rotated 90 degrees in a 3d space or such as, "what if a person moved between rooms?" Would the model understand the objects in the old room are no longer in the person's field of view?

Our benchmark wants to shift the focus towards active spatial reasoning, mental manipulation of objects, predict the outcomes of said movement, and then understand how those changes would affect its real-world use. Comparing to past benchmarks showcases a perspective on where models currently succeed and where our benchmark will provide a new test for artificial common sense.

1. Mental Rotation Focus Benchmark: SpatialViz-Bench (2026)
    - Importance: Asks models to perform "unseen relationship interference" on programmatically generated 3D objects. Results determined that the models were great at identifying 2D patterns

but struggle when a task require a mental volume in regards to 3D manipulation. Most models defaulted to making guesses based off of 2D textures rather than building a 3D mental image.

- Relevance: Our model intends to expand upon this by moving from abstract shapes to functional objects to see if common sense with help overcome this type of 3D cliff the model finds itself on.

2. Step By Step Movement Focus Benchmark: StepGame (2025)
   - Importance: Creates a story regarding an arbitrary amount of entity relations, "A is left of B, B is below C" and tests the model on the relationship between the first and last entities. The difficulty was scaled exponentially depending on how many entities were added. Sharp decline in accuracy when a relationship exceeded 3-4 steps, showing that most models lack persistent spatial memory. A sentence can be processed individually but it can't update a unified map mentally
   - Relevance: Our benchmark intends to test if a visual grounding (allowing the model to see the map while reading the steps) will fix this lapse in memory
3. Common Sense Interaction Focus Benchmark: SpaCE-Eval (2025)
   - Importance: Used over 1000 human made diagrams to test physical common sense. Questions model on environmental interaction such as "what would these two 3D shapes look like if they were combined together." Models struggle with spatial simulation, and predicting how an action changes the possibilities in the environment
   - Relevance: Common sense-esque questions target a usability gap as to whether or not a transformation makes an object or functional.

4 Structure

The benchmark will contain two broad categories of tasks:

1. Generic Transformations, as well as and "common sense" style questions as to how transformations alter the interactivity or functionality of objects.
2. Multi-Step Transformation questions, in an attempt to measure more complex spatial awareness and competence.

4.1 Generic Transformations

A series of templated spatial tasks will be presented to the model, with an answer bank containing valid answers. Models will be prompted to respond with a single word from the answer bank. Scoring is binary, questions given the correct answer from the given answer bank are correct. Incorrect answers, or answers containing words from outside of the answer bank are marked incorrect. Scoring will be displayed as total correct/incorrect, as well as by tags (2-Dimensional, 3-Dimensional, transformation only, "common sense" style question, specific object transformed). These questions will be either multi modal (image + text) or text-based, and will fit four categories:

1. 2D Spatial Transformations
2. 3D Spatial Transformations
3. 2D Spatial Transformations + Common Sense Style Question
4. 3D Spatial Transformations + Common Sense Style Question

4.1.1 Sample Tasks

4.1.2 Task Generation Strategy

Demonstrated in Figure 1 and Figure 2, 5-10 generic "template" questions will be designed, and then populated with values to create at least 50 questions. Large language models may be used to generate more questions, and if done, each question will be reviewed for accuracy and the answer will be answered manually by a team member.

Question: *Answer the following question with a one word response from the valid answers pool*
If the given image of an {arrow} is {rotated} {90} {degrees} {clockwise}, which direction is it facing?
Valid answers: up, down, left, right, forward, backward, upward, downward, clockwise, counterclockwise
Correct answer: down

Figure 1: A sample multimodal question. The image, along with words in brackets, will be swapped to create a wide variety of generic questions. Tags for scoring display would be: multimodal, 2d, transformation_only, arrow

Question: *Answer the following question with a one word response from the valid answers pool*
Using standard 3D axis nomenclature, if a {car} on a {road} is rotated about its {forward} axis {180} degrees, can it still be {driven}?
Valid answers: up, down, left, right, forward, backward, upward, downward, clockwise, counterclockwise, yes, no
Correct answer: no

Figure 2: A sample text-based question, focusing on "common-sense" usability of objects after applying transformations to their positions. Tags for scoring display would be: 3d, text_based, common_sense, car

4.2 Multi-Step Transformations

Spatial tasks that involve reasoning about a series of transformations to an object will be presented to the model. There will be two types of multi-step transformation tasks.

First, tasks where the model is given a text-based representation of the initial state of an object and a series of transformations to be performed on the object. For these tasks, the model will be prompted to respond with a text-based representation of the resulting state of the object after the transformations. Scoring is binary, only given answers that exactly match the correct answer are correct. Scoring will be displayed as total correct/incorrect.

Second, tasks where the model is given a text-based representation of the initial state of an object and a goal state which can be reached by performing transformations on the object. For these tasks, the model will be prompted to respond with the series of transformations required to reach that goal state. Options for the transformations that may be used will be provided in an answer bank and may be repeated multiple times in the output list. Scoring is based on:

1. If the answer provided produces the goal state when applied to the initial state of the object, as correct/incorrect
2. The number of transformation steps the given answer is away from the solution
3. How optimal the solution was, based on the number of transformation steps:

$$\frac{TransformationStepsTaken}{OptimalNumberofTransformationSteps} \tag{1}$$

4.2.1 Sample Task

4.2.2 Task Generation Strategy

Demonstrated in Figure 3 and Figure 4, the starting states of the 3D objects, the number of transformations being performed, and minimum number of steps required to complete the tasks will all vary. The input values will be generated by modeling these tasks in software and performing random transformations on them to ensure the states of the 3D objects are valid. Optimal solutions will also be determined using these software representations of the problems by performing breadth first search across the different transformation options until the solved state is found.

$\{Front: 2, Back: 5, Up: 6, Down: 1, Left: 3, Right: 4\}$

Question: *Answer the following question using the same format as the input.*
Given the directional faces of a numerical die and the values on those directional faces,
what are the values on the diretional faces after the following transformations?
Rotations: right face 90° clockwise, top face 90° counterclockwise, front face 180°
Correct Answer: $\{Front: 3, Back: 4, Up: 5, Down: 2, Left: 1, Right: 6\}$

Figure 3: A sample multi-step transformation question asking for the final state. The input numbered die orientation will be varied, along with types and number of rotations being performed.

Top Layer:	Bottom Layer:
[2][3]	[5][4]
[1][8]	[7][6]

Question: *Answer the following question using the same format as the input.*
Given the relative positions of the small, numbered cubes that make up a larger 2x2x2 cube
that has faces which can be rotated independently like a Rubik's Cube's, use Rubik's Cube notation to
describe rotation to describe the transformations required to get the cube into the solved state of:

Top Layer:	Bottom Layer:
[1][2]	[5][6]
[3][4]	[7][8]

Optimal Answer: $\{R', U\}$

Figure 4: A sample multi-step transformation question asking for a list of transformations. The input 2x2x2 puzzle state will be varied, along with the minimum number of steps required to solve it.